

OsdagBridge Report Generator

Technical Architecture, Modifications & Visual Engineering Report

Project: OsdagBridge (FOSSEE / Indian Institute of Technology Bombay) | Collaborator: Nidhikhare12

Author / Contributor: Aryan Vishala | Status: FINAL VERIFIED BASELINE | Tag: report-engine-production-accepted

1 Executive Summary

This report documents the technical architecture, code refactoring, and $\text{\LaTeX}$ visual rendering enhancements implemented on the **OsdagBridge Report Generator**.

The legacy reporting subsystem previously assembled PDF reports via procedural string-concatenation scripts (`chap1.py` through `chap8.py`), tightly coupling calculation outputs with raw $\text{\LaTeX}$ formatting. This approach caused margin overflows on multi-girder bridge configurations (> 5 girders), footer overlap issues, unformatted math strings, and scattered styling parameters.

The subsystem has been re-architected into a **10-Tier Semantic Report Engine**. The architecture decouples **Engineering Domain Facts**, **Document Abstract Syntax Tree (AST)**, **Centralized Styling** (`theme.py`), and **LaTeX Rendering** (`renderer.py`), backed by automated **PDF Preflight Layout Inspection** and a **350-test automated verification suite**.

- Architecture Separation Rule:
- Engineering meaning belongs in **Facts**.

• Report structure belongs in **Document AST**.

• Visual policy belongs in **Theme/Layout**.

• $\text{\LaTeX}$ generation belongs in **Renderer**.

• PDF geometry belongs in **Preflight**.

2 10-Tier Semantic Architecture Pipeline

Tier	Component	Core Responsibility	Why Introduced (Rationale)
1	Raw Payload	Captures solver results, grillage data, and inputs.	Decouples raw solver data structures.
2	Payload Validation	Fails fast on invalid span ≤ 0 , girders ≤ 1 , or missing loads.	Prevents silent crashes on malformed models.
3	Provenance Tracking	Records source dictionary keys, raw values, and unit conversions.	End-to-end data lineage; prevents unit errors.
4	ReportFacts	Immutable, strongly-typed Python domain dataclasses.	Eliminates data mutation; preserves <code>None</code> $\neq$ <code>0.0</code> .
5	Fact Validation	Asserts positive dimensions, finite demands, non-negative capacity.	Enforces domain invariants before AST building.
6	ReportDocument AST	In-memory AST (<code>Chapter</code> , <code>Section</code> , <code>Table</code> , <code>TableGroup</code> , <code>Chart</code>).	Completely separates content from $\text{\LaTeX}$ markup.
7	Structural Manifest	Computes AST fingerprints and asserts structural parity.	Tripwire preventing accidental check omissions.
8	ReportTheme	Single source of truth for page geometry, colors, table styles.	Eliminates scattered styling; instant global updates.
9	LatexRenderer	Emits <code>longtable</code> headers, <code>\cline</code> rules, and landscape flow.	Emits clean, robust $\text{\LaTeX}$ without ad-hoc strings.
10	PDFPreflight Gate	PyMuPDF binary scanner checking footer clearance and margins.	Catches physical page overflows post-compilation.

3 Scope of Work & Engineering Enhancements

Category	Requirement / Scope Item	Status	Key Technical Delivery
Core Requirement 1	Repeated Table Headers	Completed	4-tier <code>longtable</code> protocol (<code>\endfirsthead</code> , <code>\endhead</code> with <i>continued</i>)).
Core Requirement 2	Layout & Footer Overlap Fixes	Completed	Dynamic <code>\Needspace</code> clearance buffers + PyMuPDF automated footer check.
Core Requirement 3	Load & Geometry Refactoring	Completed	Separated vehicle live loads, footway, wind, seismic, and thermal tables.
Core Requirement 4	Utilization Ratio Visualizations	Completed	Figure 5.1 summary chart with $UR = 1.0$ red dashed code threshold line.
Core Requirement 5	Material Take-off Bar Charts	Completed	Figures 7.1, 7.2, 7.3 for Steel, Concrete, and Rebar with sequential numbering.
Core Requirement 6	Centralized Theme System	Completed	Dedicated <code>theme.py</code> acting as single source of truth for all layout/color rules.
Additional Enhancement	Multi-Girder Scalability	Enhanced	Verticalized Table 1 (5 columns), scaling across tested 3-, 5- and 10-girder models.
Additional Enhancement	Grouped Table Aesthetics	Enhanced	Subtle <code>\cline{2-N}</code> inner parameter dividers with full <code>\hline</code> group boundaries.
Additional Enhancement	L ^A T _E X Math AST Protection	Enhanced	AST <code>Math(...)</code> token isolation eliminating $_$ font artifacts and raw T _E X leaks.
Additional Enhancement	Preflight Safety Tripwires	Enhanced	Post-compilation automated physical PDF margin and footer collision detection.

4 Authoritative File Modification & Addition Matrix

File Path	Status	Primary Purpose & Responsibility
src/osdagbridge/core/report_engine/theme.py	NEW	Centralized styling configuration (<code>PageGeometry</code> , <code>Typography</code> , <code>ColorPalette</code> , <code>TableStyle</code>).
src/osdagbridge/core/report_engine/document.py	NEW	Intermediate AST definitions (<code>ReportDocument</code> , <code>Chapter</code> , <code>Section</code> , <code>TableGroup</code> , <code>Chart</code>).
src/osdagbridge/core/report_engine/document_builder.py	NEW	Constructs complete <code>ReportDocument</code> AST from typed <code>ReportFragments</code> .
src/osdagbridge/core/report_engine/renderer.py	NEW	Production L ^A T _E X rendering engine, math escaping, <code>longtable</code> management, and landscape section flow.
src/osdagbridge/core/report_engine/layout.py	NEW	Layout hint definitions, page break triggers, and <code>\Needspace</code> buffers.
src/osdagbridge/core/report_engine/chart_generators.py	NEW	Matplotlib chart generation service bound exclusively to <code>ReportFigure</code> objects and typography.
src/osdagbridge/core/report_engine/preflight.py	NEW	Automated PyMuPDF (<code>fitz</code>) binary PDF inspector for footer and horizontal margin verification.
src/osdagbridge/core/report_engine/manifest.py	NEW	Structural manifest extraction and legacy vs. semantic parity validation engine.
src/osdagbridge/core/report_engine/facts/	NEW	Typed domain fact models (<code>inputs.py</code> , <code>loads.py</code> , <code>design_checks.py</code> , <code>material_takeoff.py</code>).
src/osdagbridge/core/report_engine/chapters/	NEW	Semantic chapter AST builders (<code>ch2.document.py</code> , <code>ch3.document.py</code> , <code>ch5.document.py</code> , <code>ch7.document.py</code>).
src/osdagbridge/core/report_engine/validation.py	NEW	Input payload and domain fact invariant validators.
src/osdagbridge/core/report_engine/provenance.py	NEW	<code>ProvenanceTracker</code> recording raw dictionary origins and registration transformations.
src/osdagbridge/core/reports/report_generator.py	MODIFIED	Main orchestrator; integrated semantic engine, cleaned running, and linked preflight checks.
src/osdagbridge/core/reports/executive_summary.py	MODIFIED	Re-architected Table 1 into a vertical 5-column structure (<code>Girder ID Section Check UR</code>).
src/osdagbridge/core/reports/chap6.py	MODIFIED	Constrained Figure 6.1 height to 10.5cm, <code>keepaspectratio</code> to maintain view on opening page.

5 Centralized Theme Architecture (theme.py)

```
ReportTheme (Immutable Single Source of Truth)
page: PageGeometry      + A4 dimensions, 20mm margins, 25mm footer reserve
typography: TypographyStyle + 11pt Computer Modern body, 14pt headings, bold captions
colors: ColorPalette     + Osdag Brand Green (#91B014), Deep Slate Navy (#1E293B), Neutral Gray (#94A3B8)
table_styles: TableStyle + 6pt padding, inner_group_rule="subtle" (\cline), boundary_rule="strong" (\hline)
charts: ChartStyle       + 14.5cm x 5.5cm dimensions, 300 DPI, sans-serif typography
```

6 Key Technical Changes: Before vs. After Comparison

Report Element	Legacy Implementation (Before)	Semantic Report Engine (After)
Executive Summary Table 1	Horizontal columns ($G_1 \dots G_{10}$) causing right-margin overflow.	Vertical 5-column table (Girder Member Section Check UR), perfectly scalable.
Multi-Page Tables	Headers omitted on continuation pages; data disconnected across breaks.	Automatic <code>\endhead</code> repeated headers with (<i>continued</i>) captions on all pages.
Grouped Girder Tables	No internal dividing lines between parameter rows, resulting in cluttered blocks.	Subtle <code>\cline{2-N}</code> internal rules with full-width <code>\hline</code> group boundaries.
Charts & Visualizations	Default Matplotlib rainbow palette (blue, orange, teal); missing figure numbers.	Unified Osdag Brand Green (#91B014), explicit N/A badges, and auto figure numbers.
L ^A T _E X Math Formatting	Raw string escaping exposed unformatted code (<code>\times</code> , <code>\leq</code> , <code>\phi</code>) and <code>_i</code> artifacts.	Tokenized math isolation with AST <code>Math(...)</code> objects preserving mathematical typography.
Landscape Summary Flow	Default vertical stretching caused portrait page footers to float mid-page.	Isolated <code>osdaglandscape</code> environments with <code>\raggedbottom</code> and full-width header rules.
Quality Verification	Manual visual inspection of compiled PDFs.	Automated PyMuPDF <code>PDFPREFLIGHT</code> checking bounding boxes for margin/footer collisions.

7 Verification & Production Acceptance Suite

Verification Area	Primary Test File Location	Status
Content Contract	tests/contract/test_report_contract.py (11 invariants)	PASS
Structural Manifest	tests/contract/test_structural_manifest.py (0 regressions)	PASS
Data Provenance	tests/contract/test_fact_provenance.py (lineage/units)	PASS
Validation & Invariants	tests/contract/test_payload_facts_validation.py	PASS
Acceptance Matrix	tests/contract/test_acceptance_matrix.py	PASS
Production PDF Preflight	scratch/run_stage8_acceptance.py (physical PDF inspection)	PASS

Configuration Matrix	Girder Count	Physical Page Count	Footer Collision	Margin Overflow
Configuration A	3 Girders	44 Pages	PASS ($y_1 \leq 771$ pt)	PASS ($x_1 \leq 580$ pt)
Configuration B	5 Girders	51 Pages	PASS ($y_1 \leq 771$ pt)	PASS ($x_1 \leq 580$ pt)
Configuration C	10 Girders	69 Pages	PASS ($y_1 \leq 771$ pt)	PASS ($x_1 \leq 580$ pt)
Configuration D (Partial)	4 Girders	44 Pages	PASS ($y_1 \leq 771$ pt)	PASS ($x_1 \leq 580$ pt)
10-Girder Submission Report	10 Girders	82 Pages	PASS ($y_1 \leq 771$ pt)	PASS ($x_1 \leq 580$ pt)

Live Automated Test Suite Status: **350 PASSED, 1 SKIPPED, 0 FAILURES** (Execution time: 11.18s).

8 Known Limitations & Non-Goals

- Solver Boundary:** The Report Engine formats and validates data produced by OsdagBridge; it does not replace the grillage solver.
- Preflight Scope:** `PDFPREFLIGHT` verifies document layout and geometry; it is not an independent mathematical re-solver.
- Compilation Backend:** Generates standardized L^AT_EX and utilizes `pdflatex` as the production compilation backend.
- Legacy Compatibility:** The semantic engine is the production path (`REPORT_ENGINE_ENABLED = True`); legacy `chap*.py` builders are retained for manifest parity regression testing and fallback.

9 Submission Deliverables

Deliverable	Description & Contents
Report_Before.pdf	Original 10-girder baseline report (81 pages) showing legacy table overflow & formatting issues.
Report_After.pdf	Final production 10-girder report (82 pages) with vertical Table 1, repeated headers, and subtle <code>\cline</code> separators.
Technical_Changes_Report.pdf	Standalone compiled technical architecture report (3 pages) detailing code modifications and rationales.
README.md	Submission manifest, repository link, collaborator info, and test validation summary.

10 Conclusion

The OsdagBridge Semantic Report Engine has completed the defined production acceptance and regression verification process and is frozen at Git tag `report-engine-production-accepted`. The verified system is ready for submission and continued maintenance under the documented architecture.